

CSE - 6950

*The Working of ChatGPT and Ethical, Legal and Governance
Frameworks of Generative AI*

Graduate Independent Study

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Independent Study Overview

- **Goal:** understand how generative AI works technically and how to use it responsibly in the real world.
- **Spring 2025:** focused on the model side - evolution from RNNs to Transformers, pre-training, fine-tuning, RLHF, and deployment.
- **Fall 2025:** focused on the human side - law, ethics, governance, bias, transparency, and accountability.

Early Generative Models: RNNs, LSTMs & GRUs

- **Goal:** review classic sequence models that paved the way for modern generative AI.
- **RNNs:** maintain a hidden state to process sequences, but suffer from vanishing gradients and poor long-term memory.
- **LSTMs:** add input, output, and forget gates + cell state to capture long-range dependencies.
- **GRUs:** simplified LSTMs (reset & update gates) that are faster while still solving vanishing-gradient issues.

From RNNs to Transformers & Modern Generative AI

- Transition from recurrent models to Transformer architectures that use self-attention instead of recurrence.
- **Key ideas:** encoder–decoder design, self-attention, and positional encoding for order information.
- **GANs:** generator vs. discriminator in an adversarial game → high-quality synthetic images but training can be unstable
- **Diffusion models** (e.g., DALL-E, Stable Diffusion) emerge as more stable, high-fidelity alternatives for image generation.

BERT: Bidirectional Transformers for Language Understanding

- Focus on **BERT** as a bidirectional, encoder-only Transformer for deep language understanding.
- Bidirectional self-attention reads both left and right context → strong performance on QA, NLI, etc.
- Training objectives:
- **Masked Language Modeling (MLM)** – predict randomly masked tokens.
- **Next Sentence Prediction (NSP)** – learn relationships between sentence pairs.

Attention, Tokenization & Text-to-Text

- **Attention mechanism:** lets the model focus on the most relevant tokens instead of treating all words equally.
- **Self-attention** uses Query, Key, Value matrices so each token attends to all others, capturing long-range dependencies.
- Tokenization evolution: from word-level to **sub-word methods** (WordPiece/BPE) to balance vocabulary size and OOV handling

Pre-training Large Language Models

- Objective: learn broad linguistic representations from massive unlabeled corpora before task-specific tuning.
- Pre-processing pipeline: collection → cleaning & deduplication → toxicity filtering → tokenization (WordPiece/BPE).

Fine-tuning for Downstream Tasks

- **Fine-tuning:** adapt a pre-trained model to a specific task using a smaller labeled dataset → better performance with less compute.
- **Workflow:** load pre-trained model (e.g., via HuggingFace) → tokenize data → set hyperparameters → train (Trainer API) → evaluate.
- **Challenges:** overfitting on small data, catastrophic forgetting, and sensitivity to learning rate/batch size

RLHF: Workflow & Benefits

- **Reinforcement Learning with Human Feedback (RLHF):** uses human preference rankings as the reward signal to guide model behavior.
- RL fine-tuning optimizes the model to maximize predicted human satisfaction.
- **Applications:** conversational AI (ChatGPT, Claude, Gemini), safer content generation, and content moderation.
- **Benefits:** reduces harmful/toxic outputs, improves coherence, and makes responses more aligned with user expectations

AI Alignment & Challenges of RLHF

- **Goal:** align AI systems with human goals, ethical standards, and safety constraints
- **Ethical risks:** human raters bring bias; “desirable” behavior is subjective and culturally dependent
- **Reward hacking:** models may learn to please annotators rather than improve accuracy or reasoning
- **Practical issues:** high cost of human feedback, inconsistent preferences, and reward overfitting that leads to generic or overly safe replies

Inference Optimization for Deployed LLMs

- **Focus:** make inference fast, cheap, and responsive in real-time systems.
- **Objectives:** lower latency, reduce compute costs, and enable high-throughput serving for millions of users.
- **Trade-offs:** smaller or quantized models can lose accuracy; caching must handle privacy and freshness.

Scalability & User Interaction Design

- **Scalability:** ability of the system to handle more users and requests without performance degradation.
- Good UI/UX is as important as the model: bad interfaces can ruin user experience even with a strong model.
- Feedback loops from users drive continuous model improvement and safety refinement.

Legal & Ethical Dimensions of Generative AI

- Examines how tools like ChatGPT, DALL·E, Claude raise **copyright, privacy, liability, and authorship** questions.
- **Data privacy:** risk of personal/sensitive information appearing in training data or outputs.
- **Liability & ownership:** who is responsible when AI causes harm, and who owns AI-generated content?

Global AI Governance & Frameworks

- Survey of emerging frameworks: **EU AI Act**, **US AI Bill of Rights**, **UNESCO AI Ethics Recommendation**, **OECD AI Principles**.
- **UNESCO AI Ethics**: emphasizes human rights, dignity, safety, fairness, inclusion, environmental sustainability, and human-in-the-loop responsibility

Legal Accountability Foundations in Generative AI

- **Focus:** how rapid adoption of tools like ChatGPT, Claude, and Gemini outpaces clear legal rules.
- Generative AI can cause harm, but the *model* itself cannot be sued
- Responsibility falls on people and organizations: developers, deployers, and users.
- Existing laws (copyright, privacy, liability) are being stretched to cover AI behavior.

Real Cases and Legal Challenges

- Lawsuits like *New York Times v. OpenAI* question using copyrighted data to train models.
- Privacy regulators (e.g., in Italy) have fined AI firms for misusing personal data.
- Defamation cases show how hard it is to assign blame when an AI “hallucinates” false facts.

Why Terms of Use and Disclaimers Matter

- Every major AI tool is wrapped in Terms of Use and Disclaimers
- They define what users can/cannot do and warn that answers may be wrong
- These documents turn AI use into a contract that helps manage legal risk.

Comparing Major AI Providers' Terms

- OpenAI, Anthropic, and Google all say users are responsible for how they use outputs.
- Clauses limit company liability and ban harmful or illegal uses.
- There are also rules about IP ownership and “no professional legal/medical advice.”

What Is Ethical Governance of AI?

- Ethical governance means building AI that respects human rights, fairness, and safety.
- It covers the whole lifecycle: data collection, training, deployment, and monitoring.
- Key ideas: transparency, accountability, fairness, human oversight, and privacy

How Companies and Regulators Do Governance

- Companies like OpenAI, Anthropic, Google DeepMind, and Meta have ethics teams and boards.
- Global frameworks (EU AI Act, OECD, UNESCO, NIST) set high-level rules and principles.
- In practice, enforcement is uneven and sometimes conflicts with business goals.

Bias and Fairness in Large Language Models

- Bias can come from training data, model design, or human feedback.
- It shows up as stereotypes, under-representation, or unfair treatment of groups.
- Fairness audits are used to detect and reduce these biased patterns.

How Bias Audits Work (and Their Limits)

- **Common tools:** checking dataset balance, counterfactual prompts, human review, and stress tests.
- Real issues include gender roles, language inequality, and political tilt in answers
- **Challenges:** changing models, Western-centric benchmarks, and trade-offs between fairness and accuracy.

Transparency and Explainability in AI

- Full transparency is hard because models are huge and data sources are partly secret
- Companies use system cards, documentation, and policy pages to explain model behavior.
- Explainability tools (like token importance or reasoning summaries) help users see *why* an answer was given

Accountability, User Contracts, and Trust

- Accountability asks who is responsible when AI is misused or makes mistakes.
- Usage policies, logging, and compliance with laws (GDPR, CCPA, EU AI Act, NIST RMF) are key tools.
- Clear communication and good documentation help avoid over-trust and confusion about blame

Basics of Global AI Governance

- AI governance = rules and processes for building and using AI safely
- The EU AI Act uses a risk-based law with strict rules for high-risk systems.
- The US and UK rely more on sector rules and flexible principles guided by existing regulators

Role of International Bodies and Impact on Companies

- OECD and UNESCO give global ethical principles but are not binding laws.
- Companies must now do risk assessments, safety testing, and transparency reporting.
- **Goal:** protect people (privacy, fairness, safety) while still allowing innovation

What We Learned About AI Accountability

- Today's accountability tools include safety teams, audits, user policies, and legal rules
- But practice still varies a lot between countries and companies.
- **Big lessons:** laws change fast, accountability is shared, and transparency alone is not enough

Open Questions and Future Directions

- **Open questions:** who pays when AI causes harm, how transparent models should be, and whether AI outputs must be labeled.
- **Future trends:** stronger global standards, better audits, and improved explainability tools.
- Accountability is an ongoing process that must grow as AI systems evolve.

Big Takeaways & What I Learned

- Generative AI is powerful but fragile: training data and model design directly affect bias, safety, and reliability.
- Accountability is shared by developers, companies, regulators, and users all carry part of the responsibility; laws are still catching up.
- Good AI governance needs both technical tools (fairness audits, transparency, RLHF, safety checks) and strong legal/ethical frameworks